Design and Validate a Hybrid Network Using Cisco Packet Tracer

CLASS ACTIVITY 2

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Objective

Design, configure, and validate a hybrid network topology by combining two different network infrastructures. Implement either an HTTP or FTP service and demonstrate successful end-to-end communication.

Problem Statement

Design a hybrid network consisting of **two different topologies** (e.g., Star + Bus, Star + Ring, Star + Mesh, Tree + Star, etc.) using Cisco Packet Tracer.

Your network must include:

- Minimum **2 different network topologies**
- At least **8 end devices (PCs/Laptops)**
- **2 Switches**
- **1 Router**
- One **Server** configured as either:
  - HTTP Server **or**
  - FTP Server
- Appropriate cabling

Tasks

1. Design a hybrid topology by combining two different network infrastructures.
2. Create an IP addressing scheme for all devices.
3. Configure IP addresses, subnet masks, and default gateways.
4. Configure the router interfaces.
5. Configure the server:
   - Enable **HTTP** service and create a sample webpage**OR**
   - Enable **FTP** service and create a user account.
6. Verify connectivity using:
   - ping
   - tracert

7. Validate the application layer:
   - Access the webpage through a browser (HTTP)**OR**
   - Successfully log in and transfer a file using FTP.
8. Save the Packet Tracer file.

Validation Requirements

Demonstrate the following during evaluation:

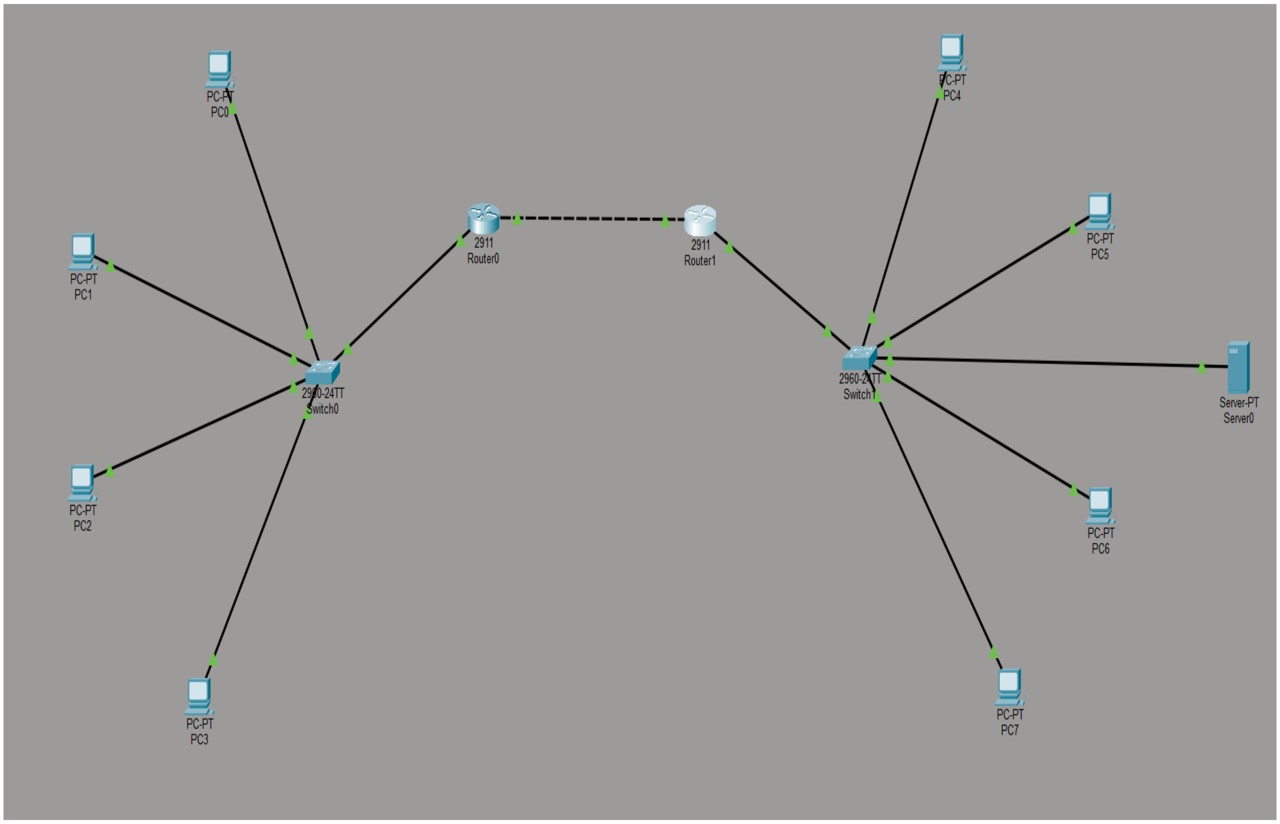
- ✔All devices receive correct IP configuration.
- ✔All hosts can communicate successfully.
- ✔No packet loss during ping.
- ✔HTTP webpage is accessible **or** FTP login and file transfer are successful.
- ✔Network is logically organized and properly labeled.

Deliverables

Submit [in single document]:

1. Network topology screenshot

Answer :



2. IP Addressing Table [table template is given below]

Answer :

IP Addressing Table:

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router 0	GigabitEthernet0/0	192.168.10.1	255.255.255.0	-
Router 0	GigabitEthernet0/1	10.0.0.1	255.255.255.252	-
Router 1	GigabitEthernet0/0	10.0.0.2	255.255.255.252	-
Router 1	GigabitEthernet0/1	192.168.20.1	255.255.255.0	-
PC0	FastEthernet0	192.168.10.2	255.255.255.0	192.168.10.1
PC1	FastEthernet0	192.168.10.3	255.255.255.0	192.168.10.1
PC2	FastEthernet0	192.168.10.4	255.255.255.0	192.168.10.1
PC3	FastEthernet0	192.168.10.5	255.255.255.0	192.168.10.1

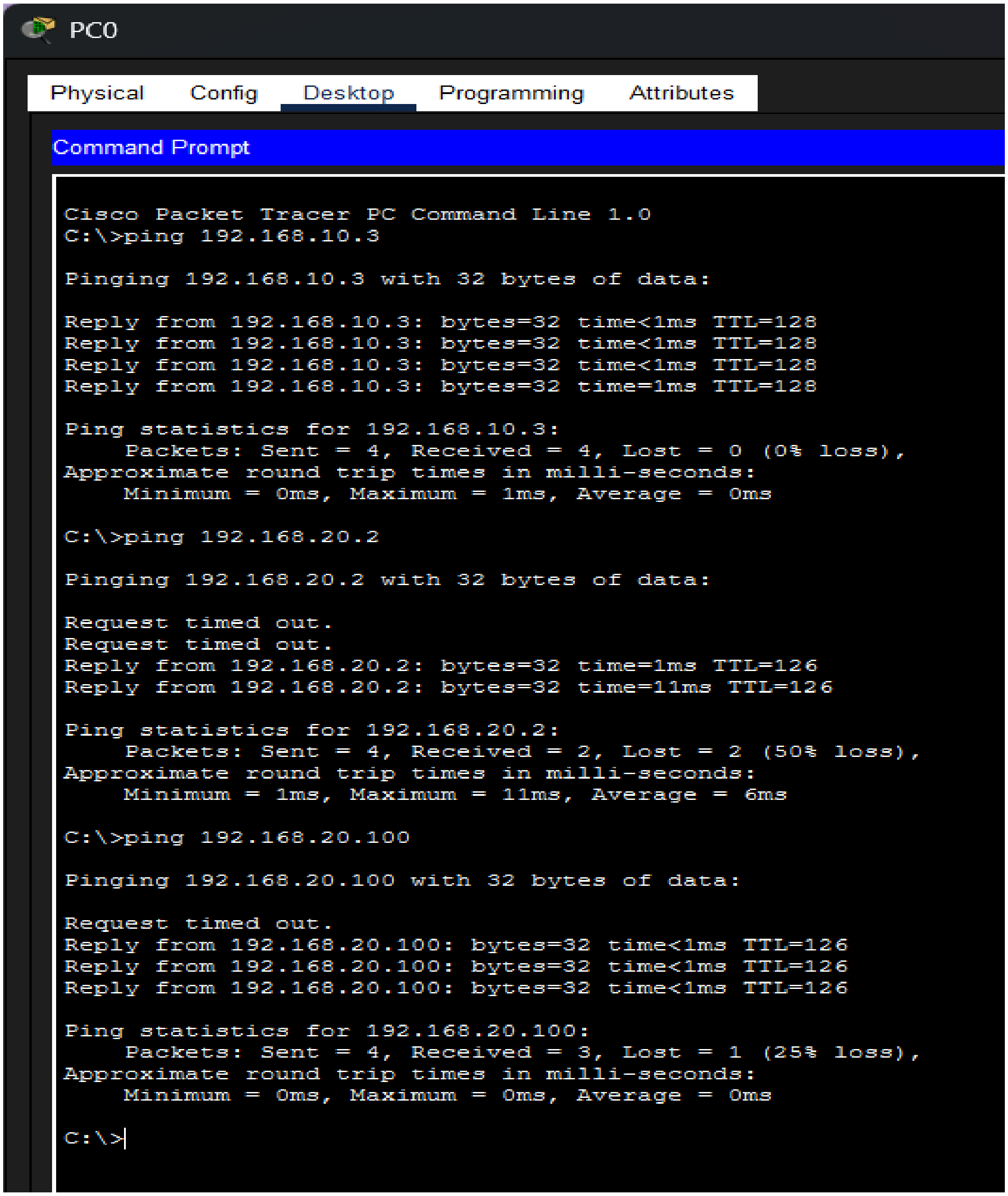
PC4	FastEthernet0	192.168.20.2	255.255.255.0	192.168.20.1
PC5	FastEthernet0	192.168.20.3	255.255.255.0	192.168.20.1
PC6	FastEthernet0	192.168.20.4	255.255.255.0	192.168.20.1
PC7	FastEthernet0	192.168.20.5	255.255.255.0	192.168.20.1
Server0	FastEthernet0	192.168.20.100	255.255.255.0	192.168.20.1

Device Interface IP Address Subnet Mask Default Gateway

1. Validation Evidence

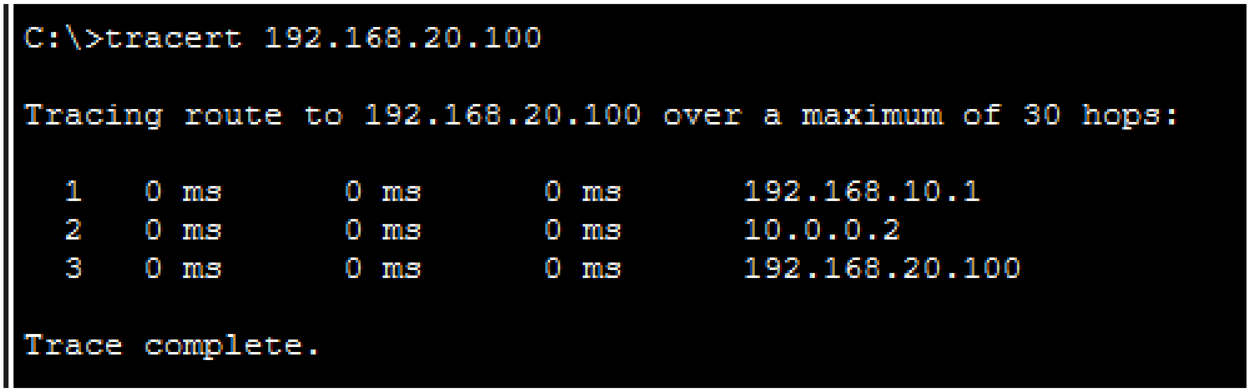
- Ping results

Answer :



- Traceroute result

Answer :



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C:\>tracert 192.168.20.100

Tracing route to 192.168.20.100 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.10.1
  2   0 ms      0 ms      0 ms      10.0.0.2
  3   0 ms      0 ms      0 ms      192.168.20.100

Trace complete.
```

- HTTP webpage screenshot or FTP login screenshot

Answer :